

LECTURE 6

Systems thinking in practice

Let's consider an example of systems thinking

Think of a food you ate today.
What systems and structures made it available for you to eat?

Consider:

- Where you purchased it
- Where it originally came from
- How it was produced and by whom
- How it was transported to the store
- Was there any waste

Those considerations have allowed us to see how different systems (agriculture, transportation, waste management, etc) are all interconnected and how they function together.

Systems thinkers understand that if one part of a system breaks down, it affects the whole system and, possibly, an entire network of systems.

When we understand the relationships between systems and their parts, we can uncover possibilities for innovations within those systems.

Causal Loop (System Thinking)

“A way of thinking about, and a language for describing and understanding, the *forces and interrelationships that shape the behavior of systems*. This discipline helps us to see how to change systems more effectively, and to act more in tune with the natural processes of the natural and economic world.” etc.

Source: The Fifth Discipline Fieldbook - Peter Senge et al

Causal Loop (System Thinking)

Two Types of Causal Loops

1. Reinforcing Loop (R) — “Snowball Effect”

A reinforcing loop amplifies change.

More leads to more, or less leads to less.

2. Balancing Loop (B) — “Stabilizer Effect”

A balancing loop tries to maintain stability or reach a target.

Activity 1: “System Mapping Sprint”

Understand boundaries, stakeholders, relationships.

Map the system of

- “Food Delivery App” or
- “Cab Booking App”

Activity 1: “System Mapping Sprint”

Instructions - Each group draws:

1. Users
2. Technology components
3. Business factors
4. Social/environmental factors
5. Arrows showing relationships

Course Project (60 + 10 Marks)

Project Duration - 4 Weeks

Team Size - 3 Members